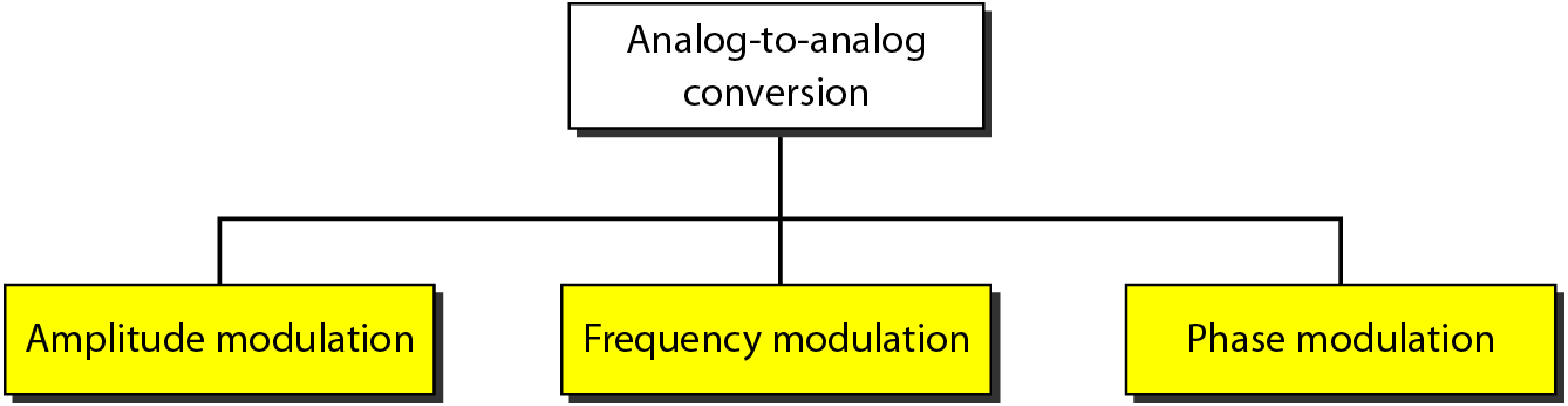
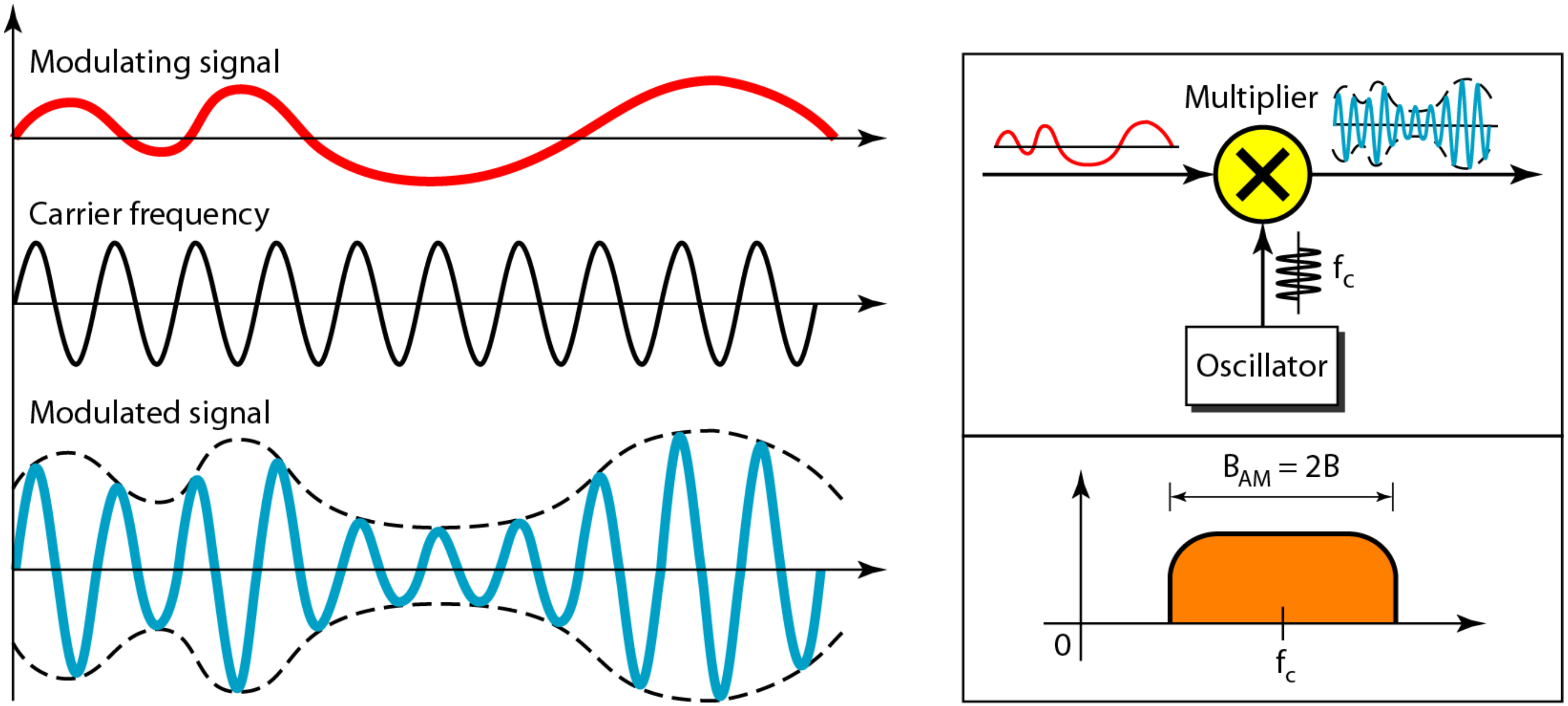
Types of analog-to-analog modulation



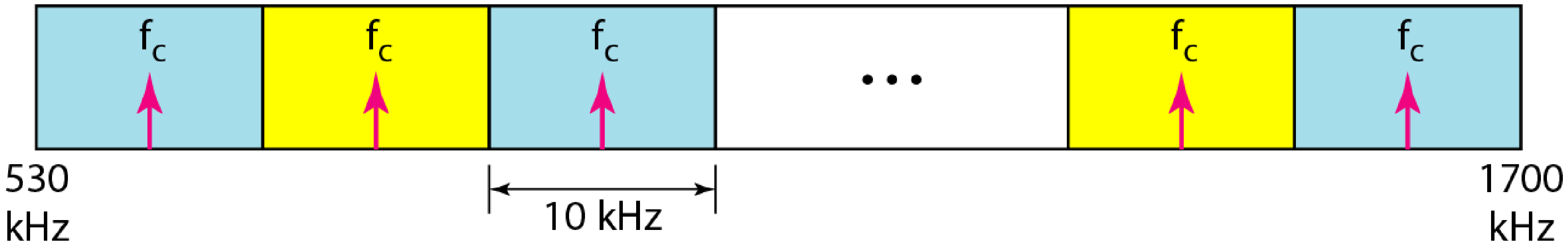
Amplitude Modulation

- **A carrier signal is modulated only in amplitude value**
- **The modulating signal is the envelope of the carrier**
- **The required bandwidth is 2B, where B is the bandwidth of the modulating signal**
- **The total bandwidth required for AM can be determined from the bandwidth of the audio signal: BAM = 2B.**
- **Bandwidth is the difference between the upper and lower frequencies in a continuous band of frequencies. It is typically measured in hertz**
- **Bandwidth of audio signal is 5kHz then an AM radio station needs bandwidth of 10kHz.**

Amplitude modulation



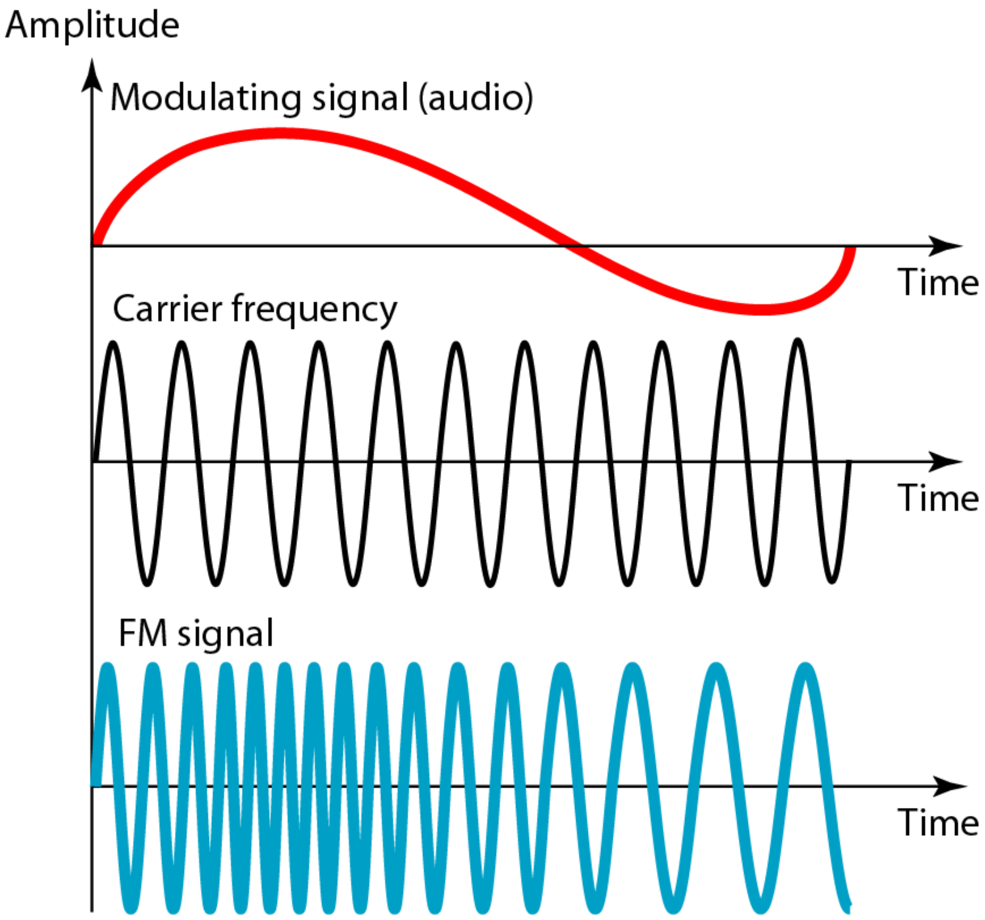
AM band allocation

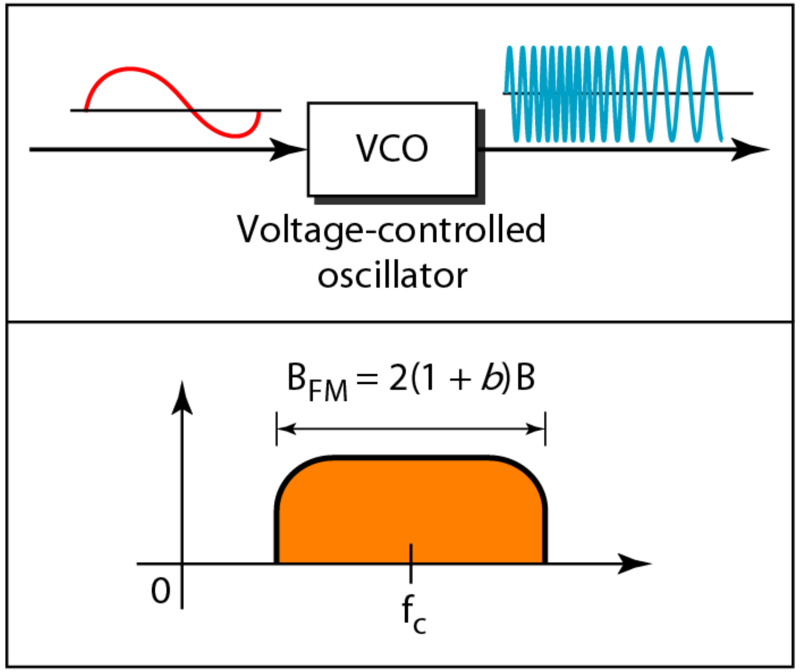


Frequency Modulation

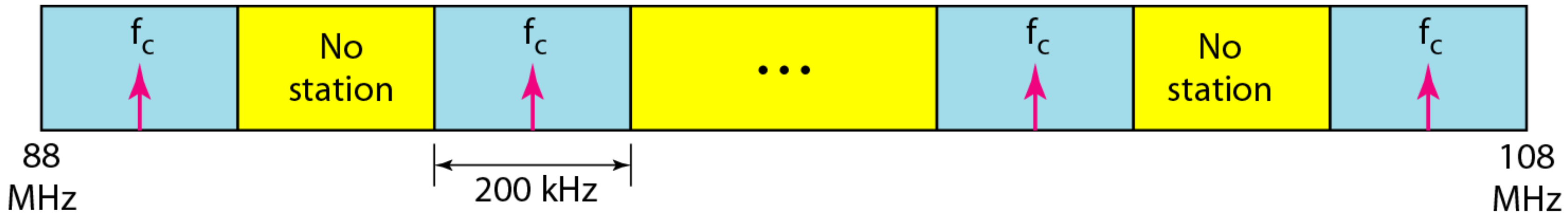
- **The modulating signal changes the freq. f_c of the carrier signal**
- **The bandwidth for FM is high**
- **It is approx. 10x the signal frequency**
- **Bandwidth of audio signal broadcast in stereo(radio) is almost 20kHz. Then the required bandwidth for each radio station is 200kHz**

Frequency modulation





FM band allocation



Phase Modulation (PM)

- **The modulating signal only changes the phase of the carrier signal.**
- **The phase change manifests itself as a frequency change but the instantaneous frequency change is proportional to the derivative of the amplitude.**
- **The bandwidth is higher than for AM.**

Phase modulation